A3-PS016-TC226

Simulation-Based Handover Decision Evaluation for High-Mobility Users

6G & Future Networks

Kondupalli Naga Mrudula • Anke Divya Dharani

Faculty Mentor: Manasa R K

Dayananda Sagar College of Engineering (DSCE), Bengaluru

AIORI Hackathon 3

October 14, 2026

Problem Description

- Users on high-speed trains and expressways (120–500 km/h) remain in a cell for only a few seconds. A late handover (HO), or a HO to the wrong cell, results in radio link failure (RLF).
- In 5G NR, the HO trigger parameters (A3 offset, hysteresis, time-to-trigger) are configured statically per cell and do not account for UE speed [1].
- Small values cause ping-pong handovers and extra signalling; large values cause too-late handovers and RLF [3].
- No simple, open tool exists to compare HO decision methods on identical scenarios using standard KPIs.

Background: How a 5G Handover Works



A3 entering condition [1]: $Mn + Ofn + Ocn - Hys > Mp + Ofp + Ocp + Off$

Small Hys / short TTT

The UE alternates between two cells (ping-pong), causing unnecessary HOs, extra signalling and data interruptions.

Large Hys / long TTT

The serving-cell signal degrades before the HO completes, leading to RLF or HO failure.

At high speed both effects are amplified: dwell time per cell is shorter and signal levels change faster. This trade-off is the focus of the project.

Literature & Standards Review

Reference	Relevance to PS-016
3GPP TS 38.331 [1]	Defines the A3 event, TTT and hysteresis used for the baseline policy
3GPP TS 38.300 [2]	Specifies the NR handover procedure and Conditional HO (CHO)
3GPP TR 36.839 [3]	Defines too-late HO, too-early HO and ping-pong KPIs used for evaluation
3GPP TR 38.901 [4]	Channel model for path loss and shadowing
3GPP TR 38.913 [5]	High-speed-train deployment scenario (up to 500 km/h)
Tayyab et al., 2019 [6]	Survey of handover management and optimisation in LTE and NR
Patriciello et al., 2019 [7]	5G-LENA (ns-3) simulator, used to cross-validate results

Proposed Solution: HO Evaluation Simulator

A modular Python simulator in which every HO method is run on the same scenario and evaluated on the same KPIs.

1 Scenario & mobility

Highway and high-speed-train routes; UE speed varied from 30 to 500 km/h through configuration files.

2 Channel & measurement

RSRP and SINR from every cell, computed with the 3GPP TR 38.901 channel model [4].

3 Pluggable HO policies

Standard A3 (baseline), speed-adaptive A3 and Conditional HO, all behind one common interface.

4 HO execution & RLF

Models HO preparation and execution delays; RLF is declared when SINR stays below threshold for T310.

5 KPI engine

Records HO count, ping-pong, RLF, HO failure, interruption time and throughput [3].

6 Results & plots

Fixed random seeds for repeatability; events logged to CSV; comparison plots per policy.

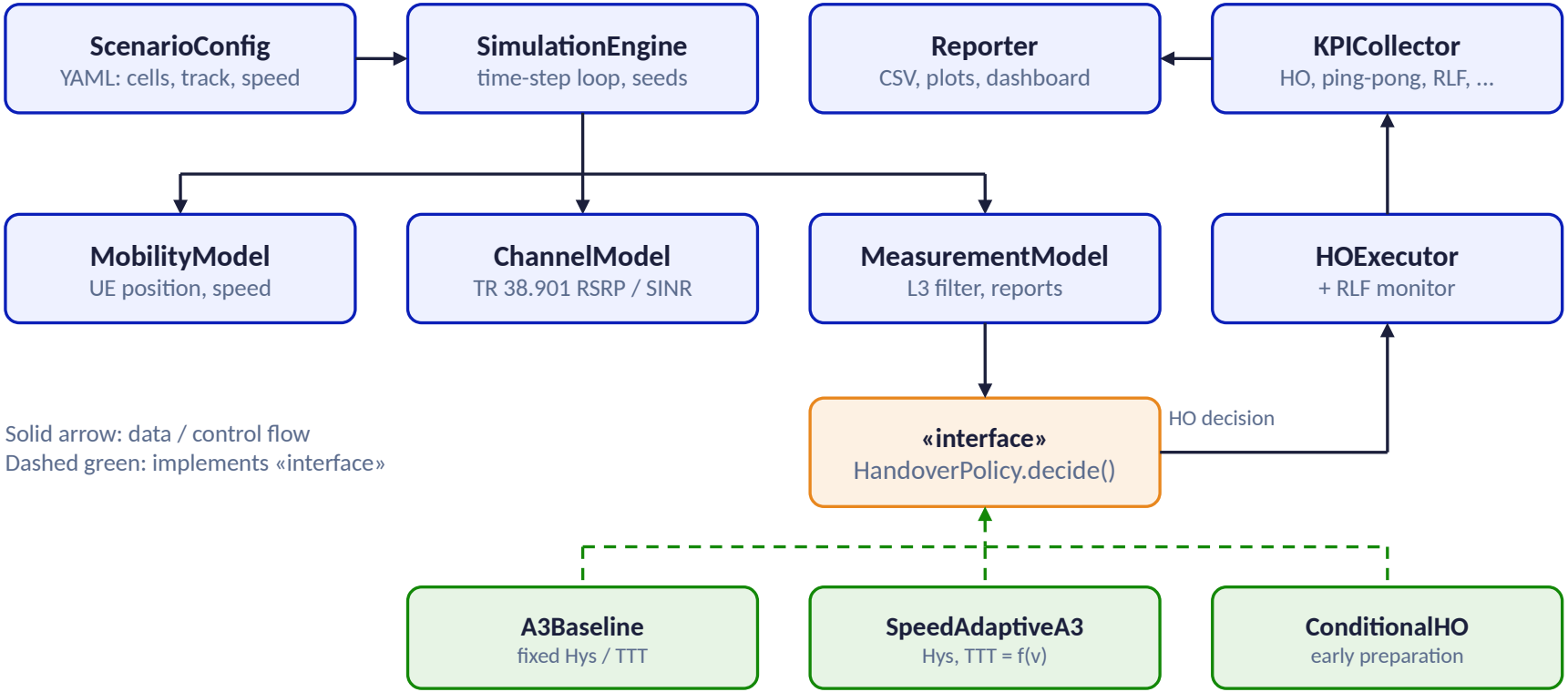
Optimization Proposed

Speed-adaptive A3: TTT and Hys are reduced as UE speed increases, so the HO is triggered in time (extends LTE speed-state scaling [8]).

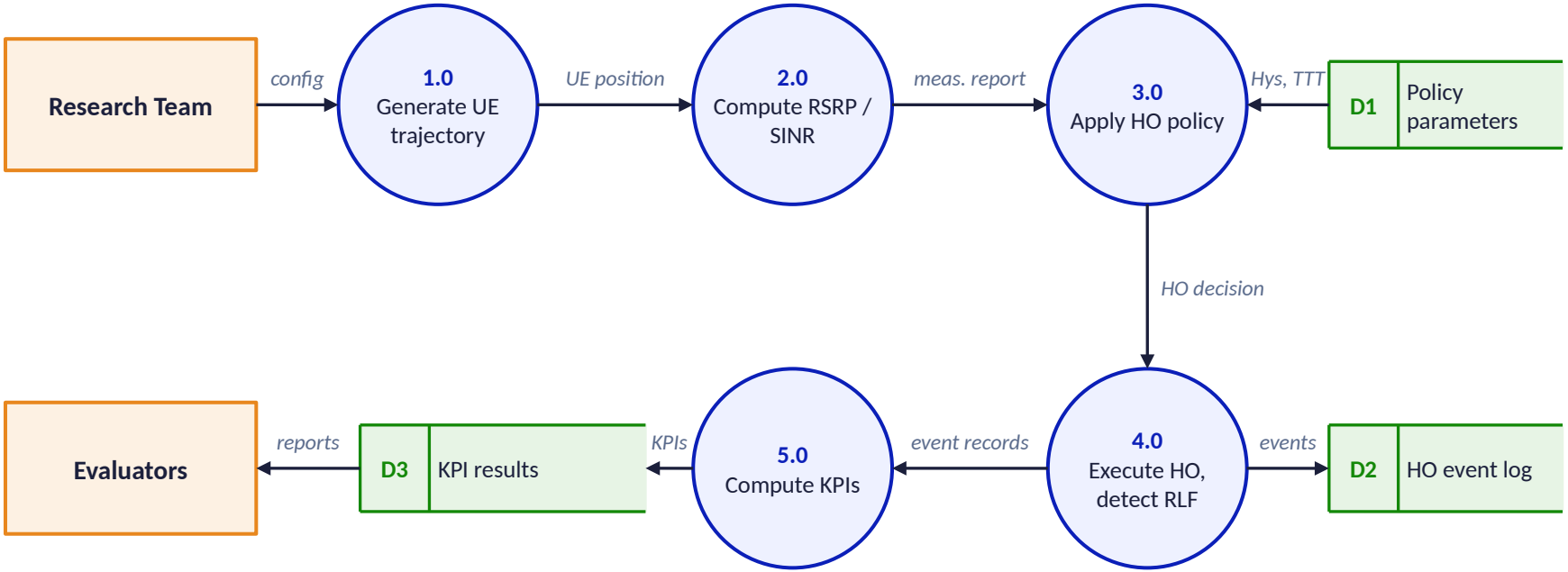
Conditional HO: candidate cells are prepared in advance and the UE executes the HO as soon as the condition is met [2].

KPI	Before: fixed A3	After: speed-adaptive A3	After: CHO + adaptive
Ping-pong rate	High at short TTT	↓ Lower	↓ Lower
Too-late HO / RLF at ≥ 300 km/h	High	↓ Lower	↓ ↓ Lowest
Unnecessary HOs	High	↓ Lower	↓ Lower
HO interruption time	Reference	≈ Same	↓ Lower
Signalling overhead	Reference	≈ Same	↑ Higher

Solution Architecture & Design



Data Flow Diagram (Level 1)



○ Process ▭ External entity ⊐ Data store → Data flow

Deployment Timeline



Methodology

1. **Baseline validation:** implement the A3 policy as per 3GPP TS 38.331 and verify HO timing against hand-calculated cases.
2. **Incremental integration:** add speed-adaptive A3 and CHO through the common HandoverPolicy interface, one policy at a time.
3. **Controlled experiments:** UE speed 30–500 km/h, Hys 0–6 dB, TTT 0–640 ms; 20 independent runs per configuration with fixed seeds.
4. **Evaluation:** compare KPIs as defined in TR 36.839, reported with mean and 95% confidence intervals.

Bibliography

[1] 3GPP TS 38.331, “NR; Radio Resource Control (RRC); Protocol specification.”

[2] 3GPP TS 38.300, “NR; NR and NG-RAN Overall description; Stage-2.”

[3] 3GPP TR 36.839, “E-UTRA; Mobility enhancements in heterogeneous networks.”

[4] 3GPP TR 38.901, “Study on channel model for frequencies from 0.5 to 100 GHz.”

[5] 3GPP TR 38.913, “Study on scenarios and requirements for next generation access technologies.”

[6] M. Tayyab, X. Gelabert, R. Jäntti, “A Survey on Handover Management: From LTE to NR,” IEEE Access, vol. 7, 2019.

[7] N. Patriciello, S. Lagén, B. Bojović, L. Giupponi, “An E2E simulator for 5G NR networks,” Simulation Modelling Practice and Theory, vol. 96, 2019.

[8] 3GPP TS 36.331, “E-UTRA; Radio Resource Control (RRC); Protocol specification” (speed-state scaling of TTT).



Thank You for Your Attention!

Team A3-PS016-TC226 • Dayananda Sagar College of Engineering, Bengaluru

Kondupalli Naga Mrudula • Anke Divya Dharani • Mentor: Manasa R K

github.com/mrudulakondupalli-pixel/A3-PS016-TC226